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Paleomagnetism and Tectonics of the Crescent Formation, Northern Olympic Mountains, Washington

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A stable prefolding magnetization has been discovered in pillow basalts of the Eocene lower Crescent Formation of the northern Olympic Mountains. The curved outcrop pattern of the Crescent Formation has been the target of several unsuccessful studies to test for oroclinal bending. The success of this study is due, in part, to the development of a small-diameter electric core drill for sampling the fractured rims of basalt pillows. Thermal demagnetization produced stable endpoints by 580°C in 12 of the 34 sites sampled (large within-site scatter was common in the remaining sites). Among the accepted sites, within-site scatter was small and correction for bedding tilt significantly reduced the scatter between sites. The mean paleomagnetic pole for newly sampled pillow basalts (86.4° north latitude, 170.0° east longitude, A_{95}=16.5°) is indistinguishable from the early to middle Eocene pole expected for North America. Including previous results from sites in subaerial basalt of the upper Crescent Formation in and near the eastern Olympic Mountains results in a more regional paleomagnetic pole (80.7° north latitude, 192.0° east longitude, A_{95}=8.0°, N=46) that shows no significant rotation (0.8° ± 14.4°) or poleward displacement (–3.6 ± 8.5). Analysis of the magnetic mineralogy suggests that the remanence is early, probably primary. The pole, therefore, should be valid for tectonic interpretation of the region. A circular distribution of virtual geomagnetic poles after correction for bedding tilt supports the hypothesis that the northern Crescent Formation experienced deformation due to the rise, in a domelike fashion, of the sediments of the Olympic Core terrane. Erosion of a partial dome open to the west could have produced the curvature seen in the outcrop pattern of the Crescent Formation. The lack of significant rotation of the northernmost Coast Ranges contrasts with the net clockwise rotation seen to the south. The difference could be that irrotational northward translation (paleomagnetically insignificant yet geologically important), driven by oblique convergence, was accommodated by the north-south trending strike-slip faults to the east of the Olympic Mountains.

INTRODUCTION

The basement of the Oregon and Washington Coast Range is characterized by Paleocene to middle Eocene tholeiitic basalts that erupted in a shallow marine environment. Whether these basalts were formed on an oceanic plate and were accreted to North America, or are the result of a rift within the western margin of North America has been controversial [*Wells et al.*, 1984; *Babcock et al.*, 1992]. Rotations of the Coast Range basalts increase from none at the north [*Beck and Engebretson*, 1982; *Irving and Massey*, 1990; this paper] to as much as 80° of net clockwise rotation to the south [*Simpson and Cox*, 1977; *Wells et al.*, 1985; *Beck*, 1989b; *Wells*, 1990] (Figure 1). *Wells and Heller* [1988] addressed the relative contribution of three proposed rotation mechanisms: rotation during accretion [*Simpson and Cox*, 1977; *Magill et al.*, 1981; *Duncan*, 1982], nonrigid dextral shear rotation due to coupling between North America and the obliquely converging Farallon or Kula plates [*Beck*, 1976, 1980; *Reidel et al.*, 1984; *Sheriff*, 1984], and rigid rotation due to differential extension of the Basin and Range Province [*Simpson and Cox*, 1977; *Magill et al.*, 1981; *Frei et al.*, 1984; *Grommé et al.*, 1986]. *Wells and Heller* [1988] concluded that a combination of rotations due to Basin and Range extension and oblique convergence fits the geologic and paleomagnetic data without invoking complicated accretion models. The lack of rotation found in the northern Coast Range, therefore, must indicate a tectonic history different from the exposures to the south.

In the Olympic Mountains of northwestern Washington, the basalts of the Coast Range belong to the Crescent Formation. Their outcrop pattern is distinctly curved (Figure 2). The curvature could be an artifact of a primary distribution of basalts, perhaps reflecting two centers of extrusion [*Cady*, 1975; *Tabor and Cady*, 1978b]. Alternatively, the curvature might have been produced by folding of a linear trend of basalts about vertical axes (oroclinal bending of *Carey* [1958]), erosion of a layered sequence deformed into an eastward plunging antiform [*Weaver*, 1937], or erosion of a domelike structure [*Cady*, 1975]. Knowledge of how this curvature formed is critical to our understanding of the tectonic processes that have been active in the northern Coast Ranges. Paleomagnetic methods should aid in resolving this structural problem. Early unpublished studies, however, found the massive lavas interbedded with pillow flows (C.S. Grommé, personal communication, 1990) of the lower (submarine) portion of the Crescent Formation to be magnetically unstable and useless for tectonic analysis.

Recent paleomagnetic studies of the upper (subaerial) portion of the Crescent Formation in and near the Olympic Mountains (Figure 2) [*Beck and Engebretson*, 1982; *Purdy*, 1987; *Warnock*, 1989] found apparent rotations, but lacked the broad geographic coverage required to unravel the regional structural and tectonic history. Magnetizations from intrusive rocks of the correlative Metchosin Igneous Complex on southern Vancouver Island (Figure 2) are concordant, but lack local structural control [*Irving and Massey*, 1990]. The goal of this project was to study the paleomagnetism of the Crescent basalts of the northern Olympic Mountains where better structural control and large areal extent might be sufficient to make sense of the existing ambiguous directions and constrain the emplacement and deformational history of the rocks of the northern Coast Range.

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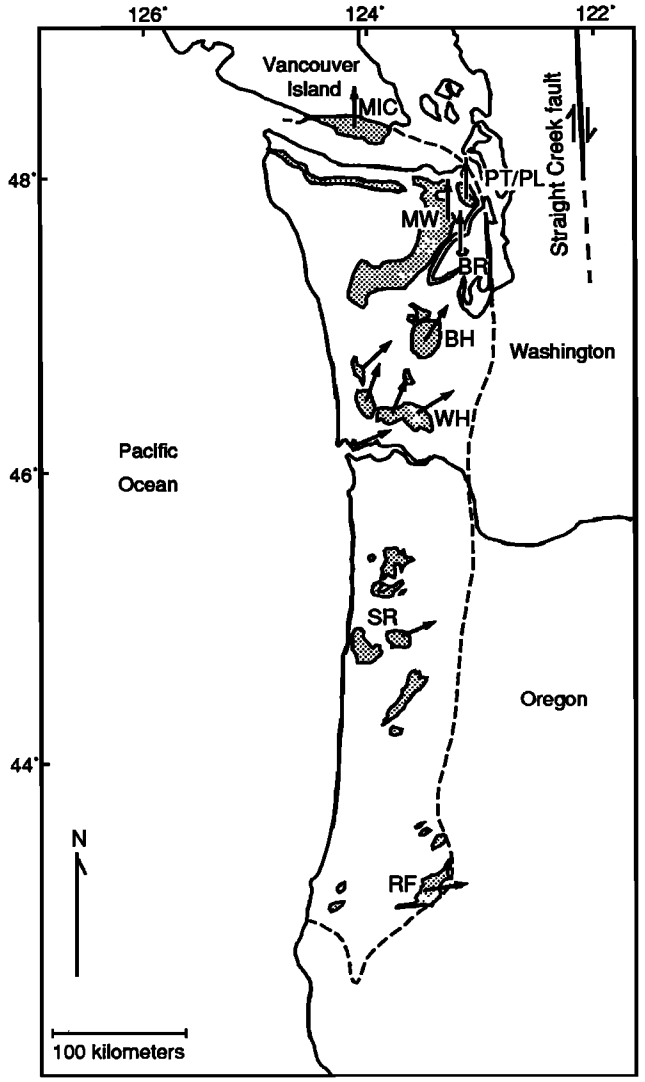


Fig. 1. Generalized geologic map of the Oregon and Washington Coast Ranges showing paleomagnetic rotations (indicated by departure of arrows from due north) in Paleocene to middle Eocene oceanic basalts (shaded regions). Note north to south increase in amount of rotation. Dashed line marks the boundary of the Coast Range Province. MIC, Metchosin Igneous Complex [*Irving and Massey*, 1990]; PT/PL, Port Townsend/Port Ludlow [*Beck and Engebretson*, 1982; *Warnock*, 1989]; MW, Mount Walker [*Warnock*, 1989]; BR, Bremerton [*Beck and Engebretson*, 1982; *Purdy*, 1987]; BH, Black Hills [*Globerman et al.*, 1982]; WH, Willipa Hills [*Wells and Coe*, 1985]; SR, Siletz River [*Simpson and Cox*, 1977]; RF, Roseburg Formation [*Wells et al.*, 1985]. Geology from *Snavely* [1987].

GEOLOGIC SETTING

The Olympic Mountains are composed of two major fault-bounded terranes: the Olympic Core terrane, and the Crescent terrane [*Silberling et al.*, 1984] (Figure 2). The Olympic Core terrane, in fault contact with the Crescent terrane on the north, east and southeast (Figure 2), represents a metamorphosed subduction complex of Eocene to Miocene turbidites [*Tabor and Cady*, 1978a]. The Crescent terrane is separated from the Olympic Core terrane by the Hurricane Ridge and Calawah faults [*Tabor and Cady*, 1978a; *Gower*, 1960] (Figure 2). The Leech River fault on Vancouver Island marks the northern boundary of the Crescent terrane [*Clowes et al.*, 1987] and the inferred "Puget fault" lies within the Puget Lowland to the east [*Johnson*, 1984, 1992; *Roberts*, 1990] (Figure 2). The Blue Mountain unit at the base of the Crescent terrane consists of continentally-derived sediments [*Cady*, 1975; *Einarsen and Engebretson*, in press]. It is overlain by and laterally interfingers with the lower member of the Crescent Formation [*Cady*, 1975], which is characterized by submarine pillow basalt with minor clastic sediments and red limestone lenses. To the east of the Olympic Core, the lower Crescent Formation is overlain by subaerial basalts of the Crescent Formation's upper member [*Glassley*, 1974]. The Crescent Formation is overlain by unmetamorphosed upper Eocene to Miocene marine clastic sediments (Figure 2). On a regional scale, the basalts of the lower Crescent Formation and the lower portion of the upper Crescent Formation appear to have been metamorphosed to the prehnite-pumpellyite facies; locally, lower Crescent basalts have been metamorphosed to the greenschist facies [*Glassley*, 1974].

Early estimates of the age of the lower Crescent Formation were made by correlation of foraminifera from intraflow limestone lenses. Foraminifera from the lower Crescent Formation indicate a lower to middle Eocene age [*Rau*, 1964; *Cady et al.*, 1972]. Five recent $^{40}Ar/^{39}Ar$ whole-rock age dates from lower and upper Crescent basalts in the Olympic Mountains range from 45.4 ± 0.6 Ma to 56.0 ± 1.0 Ma [*Babcock et al.*, 1991]. These results corroborate the fossil dates.

The Crescent Formation has been interpreted as a seamount chain that was accreted to North America [*Snavely et al.*, 1968; *Simpson and Cox*, 1977; *Duncan*, 1982]. However, there are inconsistencies with this interpretation. The interfingering of continentally derived sediments of the Blue Mountain unit at the base of the Crescent terrane, three-meter diameter clasts of 65 million year old quartz diorite within the eastern section of the Crescent Formation, and the conformable marine clastic sedimentary units overlying the Crescent Formation suggest a near-shore setting [*Cady*, 1975]. Recent structural and stratigraphic studies of the Metchosin Igneous Complex [*Brandon and Massey*, 1985; *Massey*, 1986] and similar studies including the geochemistry of the basalts of the eastern section of the Crescent Formation [*Clark*, 1989; *Babcock et al.*, 1992] suggest that a continental margin rift setting [*Wells et al.*, 1984] may be a more appropriate interpretation. Marginal rifting is consistent with plate reconstructions that show the Kula and Farallon plates interacting with northwestern North America at the time of extrusion [*Engebretson et al.*, 1985; *Stock and Molnar*, 1988]. The interaction of the Kula-Farallon ridge with the North American subduction zone suggests that a "no-slab window" may have been partially responsible for rifting of the continental margin [*Babcock et al.*, 1992].

PALEOMAGNETISM

Sample Collection and Determination of Paleohorizontal

Five to fourteen (mostly six to eight) cores were collected from each of 33 sites of pillow basalt of the lower Crescent Formation in the northern Olympic Mountains. Sampling was restricted to roadcuts in order to obtain the freshest rock possible for magnetic analysis, and to cover the largest area with the time and resources available. Suitable roadcuts from the Dosewallips River (southeast) to Ellis Mountain (northwest) were sampled (Figure 2). Active logging on roads with highly-weathered outcrops discouraged sampling west of Ellis Mountain.

Six sites near Lake Sutherland (LS in Figure 2) were sampled with a standard 2.54 cm diameter core drill. Remaining sites were drilled with a specially-designed, battery-operated drill and small bit (0.9 cm inside diameter) in order to precisely sample fractured pillow rims. Rims were targeted because a pilot study showed

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that rims are more likely than pillow cores to record a stable magnetization (Figure 3).

Most samples were oriented in situ using a sun compass and checked with a magnetic compass. In the absence of sun, a second compass held away from the outcrop verified the absence of detrimental magnetic gradients. The 2.54 and 0.9 cm diameter cores were cut into specimens 2.25 and 0.8 cm in length, respectively.

Pillows selected had spherical to slightly oblate morphologies, which typically develop on horizontal or gently-dipping surfaces [*Ballard and Moore*, 1977]. Orientations of pillow keels and sedimentary interbeds agreed and were used to obtain bedding attitudes and top directions.

Measurement and Demagnetization of NRM

Magnetic remanence was measured on a Schonstedt SSM-1a spinner magnetometer. Specimens with inhomogeneous magnetizations (γ_{95} greater than or equal to 15° [*Briden and Arthur*, 1981]) were rejected. Specimens had natural remanent magnetization (NRM) moments per unit volume ranging from 10^{-2} to 10^{2} A/m. The majority of specimens fell within the range of 10^{-1} to 1 A/m. One site, having a magnetization that was weaker than the noise level of the magnetometer, yielded no useful results.

To test stability and determine the unblocking temperature spectra of the magnetizations, two pilot specimens (mostly from

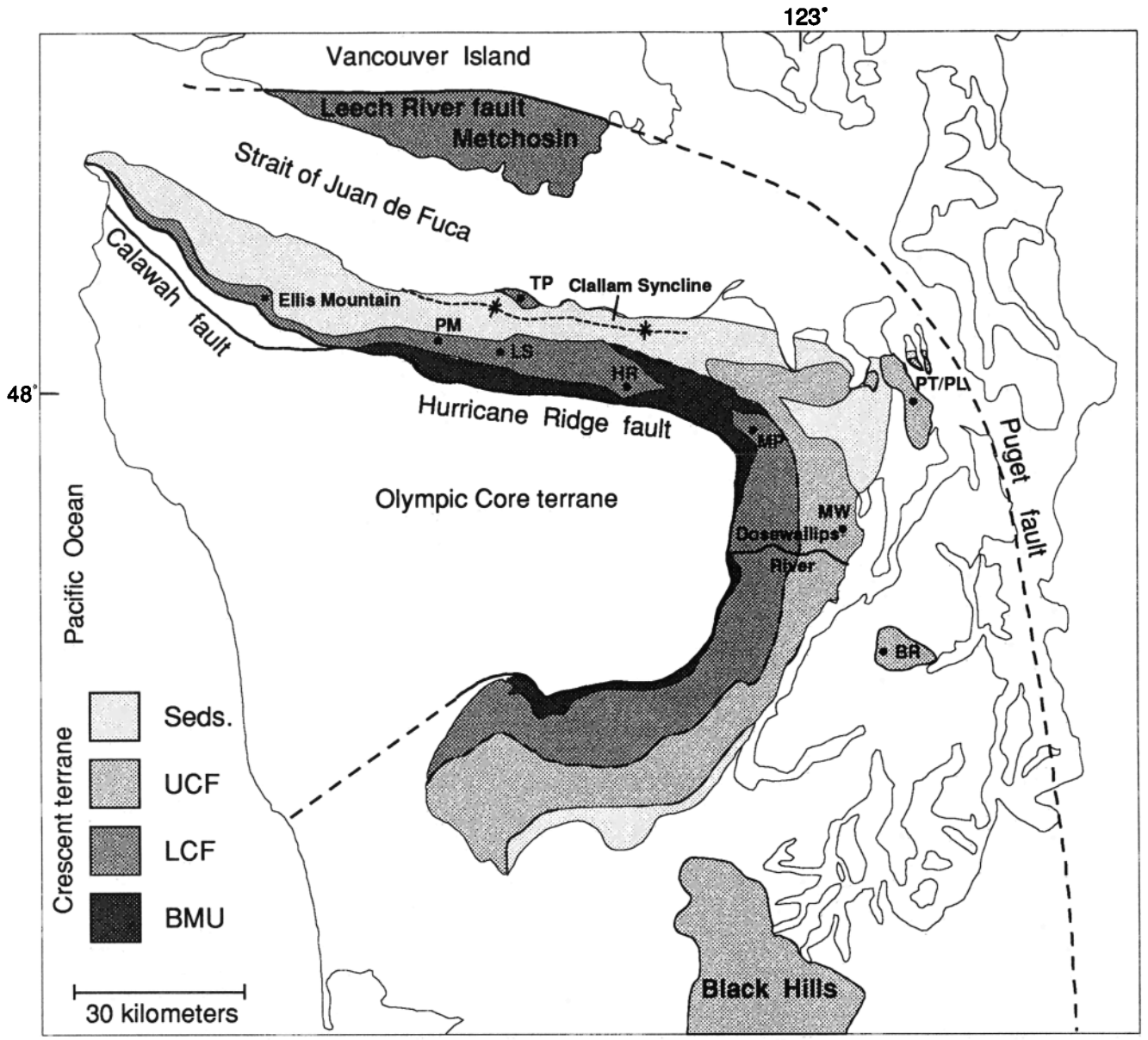


Fig. 2. Generalized geologic map of the Olympic Peninsula showing the curvature of the Crescent Formation outcrop pattern and paleomagnetic sampling localities. PM, Pyramid Mountain; LS, Lake Sutherland; HR, Hurricane Ridge; and MP, Maynard Peak (this study), TP, Tongue Point [*Moyer*, 1985]; MW, Mount Walker [*Warnock*, 1989]; PT/PL, Port Townsend/Port Ludlow [*Beck and Engebretson*, 1982; *Warnock*, 1989]; BR, Bremerton [*Beck and Engebretson*, 1982; *Purdy*, 1987]. Seds, sedimentary rocks (upper Eocene to Miocene); UCF, upper Crescent Formation; LCF, lower Crescent Formation; BMU, Blue Mountain unit. Geology from *Tabor and Cady* [1978a].

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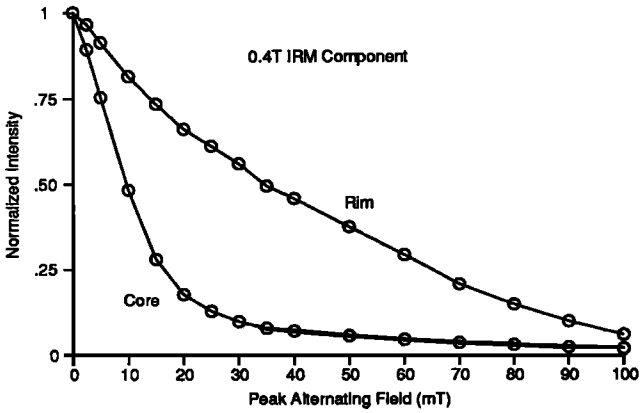


Fig. 3. Stability of a 0.4-T isothermal remanent magnetization (IRM) to alternating field demagnetization for the rim and core of a single pillow. Samples were first given an IRM in a 1-T field down the length of the specimen (z axis), then in a 0.4-T field perpendicular to the z axis. This was done to isolate the component of magnetization which had coercivities matching those of the natural remanent magnetization. This diagram shows only the component that carries the 0.4-T IRM. The rim shows a stable, nearly linear decay in magnetization with increased peak alternating field. The core, however, shows a rapid decrease by 20 mT indicating that the remanence is carried primarily by low coercivity (unstable) minerals.

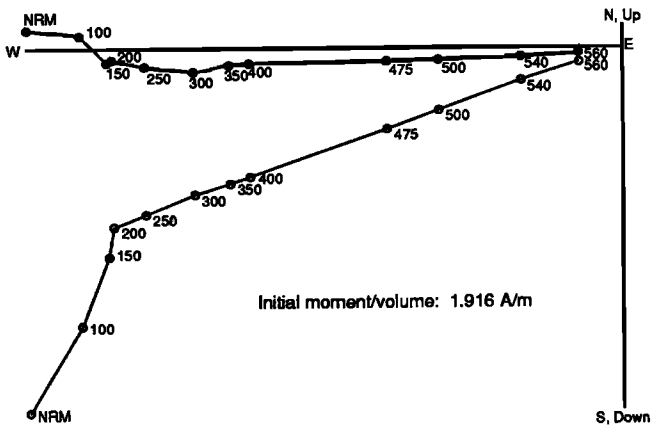


Fig. 4. Orthogonal projection of the thermal demagnetization path for specimen PM010101 showing removal of an overprint by approximately 300°C. The free line method of principal component analysis [*Kirschvink*, 1980] allows the extraction of portions (in this case 350° to 560°C) of the demagnetization path without regard to other components that may be present. Solid circles, projection onto horizontal plane; open circles, projection onto vertical west-east plane.

the same core) from each site were subjected to two different types of progressive demagnetization: alternating field (AF) and thermal. The AF pilot specimens were demagnetized in a Schonstedt GSD-5 tumbling specimen demagnetizer at 15 levels between 10 and 100 mT. Thermal pilots were demagnetized in air at 13 levels between 100° and 580°C in a custom-made, magnetically shielded oven. Oven calibration experiments established that actual temperatures are no more than 10° below the set point. Before initial heating, and after each subsequent step, magnetic susceptibility of each specimen was measured to monitor thermally-induced changes in magnetic mineralogy. Because it appeared that the bulk of the magnetization was carried by a relatively high coercivity, fine-grained, Ti-poor titanomagnetite, all remaining specimens were thermally demagnetized with seven steps from 300° to 585°C.

Analysis of NRM Demagnetization

Results of progressive demagnetization were plotted as standard orthogonal projections [*Zijderveld*, 1967]. Most specimens showed two linear segments, one of which is interpreted as a relatively weak overprint removed by 200° to 300°C (Figure 4). Directions of the more stable components were determined using the free-line method of principal component analysis [*Kirschvink*, 1980]. A maximum angular deviation (MAD of *Kirschvink* [1980]) of 15° for such line fits was used to filter out samples with unacceptably nonlinear components.

The distribution of sample directions within each site was checked for circularity at the 95% confidence level using a method outlined by *Schmidt* [1990] that incorporates principal component analysis. This was an aid in identifying sites possibly biased by a recent magnetization. With one exception, all sites that passed subsequent confidence tests had distributions that were circular at the 95% confidence level (Figure 5). Mean directions for all sites were calculated using the method of *Fisher* [1953]. Sites were excluded if they had unusually large angular deviations (MAD > 15°) or mean directions with uselessly large confidence regions (α_{95} > 15°). Accepted site-mean directions were converted into virtual geomagnetic poles (VGPs) for comparison with reference poles.

Structural correction of the site-mean directions involved rotating the in situ remanence directions to paleohorizontal about the local strike. The in situ directions also were corrected for fold plunge, then residual tilt to test *Weaver's* [1937] hypothesis that the Crescent Formation's curved exposure resulted from erosion of a plunging fold. The azimuth (90°) and plunge (45°) of the possible fold axis were approximated by fitting a great circle to

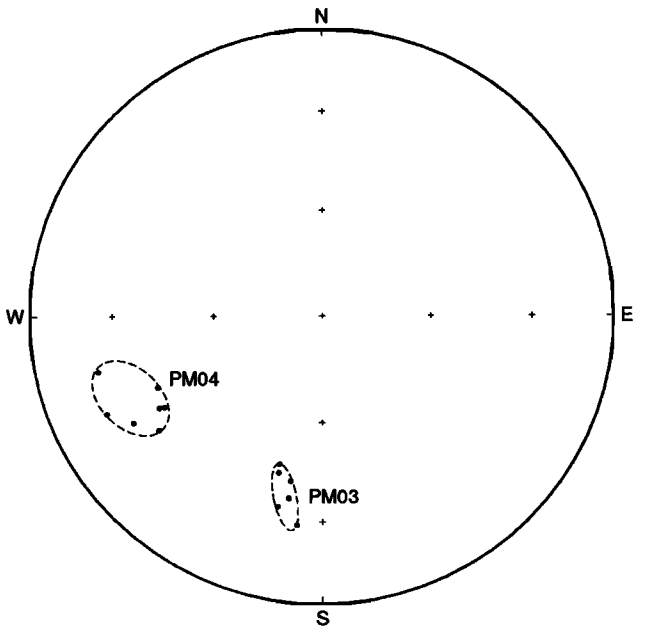


Fig. 5. Equal-area projection of sample directions for two sites. Site PM03 is the only site for which the hypothesis that the samples were drawn from a circular distribution can be rejected at the 95% confidence level [*Schmidt*, 1990]. This apparent noncircularity may be due to minor differential rotations within the site or variable contributions from a minor stable secondary component of magnetization. Note, however, that the distribution along the short axis is smaller than typical, rather than the scatter parallel with the long axis being enlarged as might be expected of mixed components. All other sites, for example PM04, may appear elliptical, but due to the small number of sample directions the assumption that the distribution is circular cannot be rejected at the 95% confidence level.

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widely scattered attitude data as measured in the field and taken from the map of *Tabor and Cady* [1978a]. Only measurements with top indicators were used.

Paleomagnetic Results

Of the 14 sites that passed confidence screenings, two sites (LS01 and PM06) were excluded from this study on the suspicion that they were sampled from recently rotated blocks. Site LS01 was subsequently found to be located within the large Mount Storm King landslide [*Logan and Schuster*, 1991] and site PM06 had a strike oriented at a right angle to the strikes of adjacent sites. Site HR03 was excluded on suspicion of being completely remagnetized; its uncorrected direction was that of the present-day field. Site TP01, sampled by *Moyer* [1985], was included because it is Crescent basalt and techniques similar to those of this study were used for its analysis.

Paleomagnetic and structural data are summarized in Table 1. Using the test of *McElhinny* [1964], the reduction in scatter on correction for tilt (k increases from 3.1 to 16.4) is significant at the 95% confidence level (Figure 6). Incorporation of fold plunge in the correction produces a slightly larger scatter (k=14.3). Localities LS and MP have upward directions after correction, indicating that acquisition of magnetization spanned at least one reversal of the dipole field. Differences of site-mean directions obtained from adjacent flows were verified as significant using the test of *McFadden and Lowes* [1981]. Two sites at the HR locality had indistinguishable directions at the 95% confidence level. This similarity appears to be fortuitous because they are stratigraphically separated by two sites with significantly different directions of magnetization.

VGPs calculated from tilt-corrected site-mean directions are circularly distributed at the 95% confidence level [*Schmidt*, 1990] but show large (27°) dispersion; this will be discussed in the next section. The paleomagnetic pole obtained by averaging site-mean VGPs (86.4° north latitude, 170.0° east longitude, and A_{95}=16.5°) is virtually identical to the 44-54 Ma craton pole of *Diehl et al.* [1983] (Table 3).

Comparison with Previous Paleomagnetic Studies

Results from this study were compared to those of previous work on the upper Crescent Formation from localities near Bremerton (BR) [*Beck and Engebretson*, 1982; *Purdy*, 1987], Port Townsend and Port Ludlow (PT/PL) [*Beck and Engebretson*, 1982; *Warnock*, 1989], and Mount Walker (MW) [*Warnock*, 1989] (Figure 2 and Tables 2 and 3). Combined, these previously studied exposures and those of the lower Crescent (present study) comprise four structural domains. Comparison of results from these domains is facilitated because the basalts of the Crescent Formation were extruded and magnetized during the Eocene when there was little apparent polar wander [*Diehl et al.*, 1983]. Since each domain is essentially homoclinal, each was used as a limb in a fold test [*McFadden and Jones*, 1981] to determine whether the lower and upper Crescent Formation record a common dipole field. At the 95% confidence level, the four domains pass the fold test (Figure 7). Normal and reverse (after inversion) sites share a common mean, according to the test of *McFadden and Lowes* [1981].

The observed 27° of dispersion in VGPs from this study and the 28° in the combined data set is larger than the 18° to 20° one might expect for 48°N latitude during the Eocene from a simple model of paleosecular variation based on global compilations of other results [*McFadden and McElhinny*, 1984]. However, these compilations and the model of *McFadden and McElhinny* [1984] reflect VGP distributions that have been arbitrarily screened to exclude poles that lie at low to middle latitudes; this may explain the observed discrepancy [*Kristjansson and Johannesson*, 1989]. Comparison with other paleomagnetic studies of the Oregon and Washington Coast Range basalts [e.g., *Simpson and Cox*, 1977; *Globerman et al.*, 1982; *Beck and Engebretson*, 1982; *Wells and Coe*, 1985] reveals that large dispersions are typical.

Averaging the site-mean VGPs of all four domains appears to provide a good approximation of the paleomagnetic pole for the northern and north-eastern sections of the Crescent Formation. Comparison of this mean pole with the pole from cratonal North America shows no significant discordance (Table 3 and Figure 7).

Timing of Acquisition of Magnetization

Because a predeformational magnetization is not necessarily a primary magnetization [*Burmester et al.*, 1990], a detailed analysis of the magnetic minerals is required in order to fully unravel the paleomagnetic history of the region. Geochemical results from the lower Crescent Formation indicate that their TiO_2 content is slightly higher than expected for mid-oceanic ridge basalts [*Babcock et al.*, 1992]. This high TiO_2 content and observations from fresh pillows dredged from active mid-oceanic ridges [*Irving*, 1970; *Irving et al.*, 1970] support the assumption that the primary magnetic phase probably was titanomagnetite. In addition, observation of polished thin sections from a pillow core reveals skeletal growth crystal morphologies (mostly cruciform) typical of rapidly quenched titanomagnetite (Figure 8). Under conditions typical of low temperature seafloor alteration, titanomagnetite readily oxidizes to a metastable titanomaghemite [*Irving*, 1970].

The present magnetic mineral, however, is not titanomagnetite or titanomaghemite. Results of two Curie temperature measurements for the rim and core of a single, typical pillow are shown in Figure 9. These thermomagnetic curves display a behavior characteristic of pure magnetite or Ti-poor titanomagnetite with a possible minor amount of pyrrhotite in the rim of the pillow. The pyrrhotite may be the result of a reducing environment either during cooling or reheating [*Irving et al.*, 1970], whereas magnetite is likely the result of the inversion of titanomaghemite caused by reheating. Seafloor metamorphism [*Alt et al.*, 1986] to the prehnite-pumpellyite facies may best explain the metamorphism of the lower Crescent Formation, however, burial metamorphism [*Coombs et al.*, 1959] is also possible given the presence of sedimentary rocks overlying the northern and eastern sections of the Crescent Formation. Heating between 250° and 300°C (the approximate range for the prehnite-pumpellyite facies of metamorphism [*Liou*, 1971]) causes titanomaghemite to invert to ilmenite/anatase and a titanomagnetite which is magnetically indistinguishable from pure magnetite [*Ozdemir*, 1987]. Optical examination of thin sections reveals an abundance of fine-grained reddish material (possibly anatase) which appears to be associated with the skeletal titanomagnetite. The remanence observed, therefore, is most likely a chemical remanent magnetization. However, we find the occurrence of distinct site-mean directions in the lower Crescent, even within homoclinal localities, to favor early remagnetization, or inheritance of a primary magnetization by the magnetite during the inversion process.

Subaerial basalts of the upper Crescent Formation appear to have a primary magnetization; polished thin sections reveal a texture (Figure 10) that is characteristic of extreme deuteric oxidation [*Haggerty*, 1991]. High oxygen fugacity conditions during cooling cause primary Ti-rich titanomagnetite to oxidize to Ti-poor titanomagnetite with ilmenite lamellae, and primary Ti-rich titanohematite to hematite and pseudobrookite [*Grommé et al.*,

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TABLE 1. Paleomagnetic Site-Mean Directions and Structural Data for the Lower Crescent Formation

	Uncorrected		Corrected							Location	
Site	Dec	Inc	Dec	Inc	N	R	α_{95}	DipAz	Dip	Lat	Lon
HR01	152.1	38.8	16.7	68.7	3	2.996	5.7	346	69	47.99	236.62
HR02	174.1	51.6	357.3	64.5	5	4.901	12.2	355	64	47.99	236.62
HR04	168.7	49.1	21.1	48.6	4	3.944	12.7	5	80	47.99	236.62
LS02	69.3	15.9	149.4	-49.6	3	2.981	12.2	32	131	48.05	236.30
LS06	61.2	30.3	137.3	-64.5	4	3.962	10.4	32	131	48.05	236.30
MP01	29.5	-71.3	221.4	-46.5	7	6.827	10.4	38	62	47.92	236.89
MP02	64.3	-2.6	181.2	-85.0	8	7.977	3.2	60	90	47.91	236.89
PM01	259.5	20.8	286.1	56.3	7	6.913	7.3	50	46	48.07	236.19
PM02	218.6	32.3	319.6	82.3	6	5.974	4.9	30	60	48.08	236.14
PM03	191.7	37.9	6.4	68.0	6	5.957	6.2	10	74	48.08	236.14
PM04	243.3	31.6	348.9	49.0	7	6.919	7.1	32	87	48.08	236.09
TP01	343.0	43.5	335.4	54.1	16	15.599	6.2	190	13	48.16	236.29
Mean	225.2	37.3	-	-	12	8.402	30.0	-	-	48.00	236.40
Mean	-	-	350.4	65.2	12	11.33	11.1	-	-	48.00	236.40

Dec and Inc, declination and inclination of site-mean directions; N, number of accepted samples; R, length of resultant vector; α_{95}, half-angle of cone of 95% confidence; DipAz, azimuth of downdip vector; Dip, angle of bedding tilt (angles greater than 90° indicate overturned beds); Lat and Lon, north latitude and east longitude of sampling site.

1969]. Rapid oxidation would explain why the Ti-poor phase (magnetite) and hematite share a single component of magnetization (Figure 11). The magnetization should have been unaffected by reheating because the oxidation that took place early resulted in a higher Curie temperature for (Ti-poor) titanomagnetite (580°C). Moreover, the presence of a reverse direction, sandwiched between normally magnetized flows at the MW locality, is difficult to explain by late remagnetization.

The successful multilimb fold and reversal tests (Figure 7) including both the upper and lower Crescent Formation indicate that the submarine and subaerial flows recorded a common dipole field before deformation. In spite of the low-grade metamorphism of the lower Crescent basalts, the magnetization appears to be early, if not primary. The paleomagnetic pole determined from the combined data set should be valid for the tectonic analysis of the region which follows.

REGIONAL TECTONICS

Deformation of the Crescent Formation

Proposed models for the deformation of the Crescent Formation in the Olympic Mountains can be tested using paleomagne-

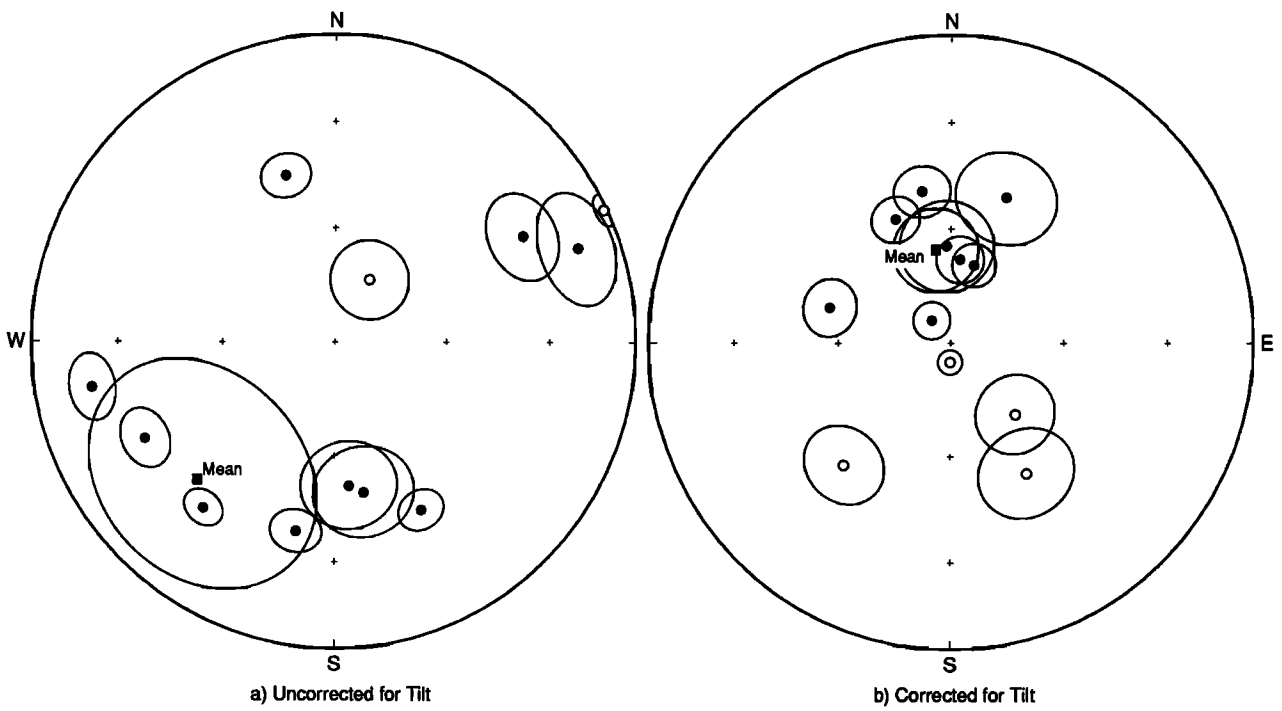


Fig. 6. Equal-area projections of site-mean directions for the lower Crescent Formation (a) before and (b) after correction for bedding tilt. Reduction in scatter after correction for bedding tilt is significant at the 95% confidence level [*McElhinny*, 1964]. Closed symbols indicate downward directions and open symbols represent upward directions. Ellipses are α_{95}.

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tism. The oroclinal bending model of *Carey* [1958] involves the folding of an originally linear trend of basalts about vertical axes. The predicted fanning of paleomagnetic declinations (Figure 12a) is not observed in the northern Olympic Mountains; therefore this model is dismissed. However, it should be noted that Olympic exposures south of Mount Walker (Figure 2) have not been sampled and remain unconstrained paleomagnetically.

Erosion of a layered sequence deformed into an eastward plunging antiform [*Weaver*, 1937] could have produced the observed curvature. If this were the appropriate model, structural correction solely for bedding tilt would have resulted in false rotations due to failure to correct for plunge [*MacDonald*, 1980] (Figure 12b). Not only is no such pattern evident in the data, but when the data are corrected for plunge, then residual tilt, directions are slightly more scattered than after simple tilt correction. Therefore, this model also is not preferred.

A simple domal uplift rotates beds about horizontal axes (strike). Erosion of a domelike structure would result in a curved outcrop pattern [*Cady*, 1975]. Because correction of Crescent site-mean directions for bedding tilts about strike produces a circular distribution of VGPs, the simplest model that describes the deformation and is consistent with the paleomagnetic data is a domelike uplift. The outcrop pattern of the Crescent Formation, however, does not display a complete circular dome, but is open to the west and southwest. An explanation for this configuration is that the exposure pattern reflects the original distribution of basalts; the northern and eastern sections may therefore represent two centers of extrusion [*Cady*, 1975] (Figure 12c).

TABLE 2. Site-Mean Virtual Geomagnetic Poles for the Crescent Formation

		Uncorrected		Corrected	
Site	N	North Latitude	East Longitude	North Latitude	East Longitude
PL01	9	-80.3	345.6	-45.2	311.6
PL03	6	-64.1	11.5	-59.7	285.1
PT01	7	-81.7	260.6	-73.0	208.1
PT02	6	-69.4	230.9	-62.1	215.8
PT03	6	-6.1	24.0	-57.6	15.9
PT04	5	-0.2	52.1	-50.7	79.1
PT05	6	-16.5	11.6	-61.3	337.5
PT06	6	-13.7	9.9	-34.4	344.8
PT07	8	-31.9	119.8	-44.5	114.4
PT08	9	-57.1	43.9	-53.6	345.1
PT09	8	1.2	34.0	-40.0	17.1
BR01	6	-62.5	209.7	-84.1	175.9
BR02	6	-66.8	211.5	-86.9	116.1
BR03	5	-77.3	161.3	-72.5	80.6
BR04	4	-51.6	168.1	-60.7	178.0
BR05	7	-70.2	150.9	-86.0	140.6
BR06	3	-70.6	138.5	-83.5	93.0
BR07	6	-56.4	165.1	-66.9	175.8
BR08	9	-72.9	23.0	-72.9	23.0
BR09	4	-59.1	145.9	-75.0	89.2
BR10	5	-58.1	157.4	-80.7	105.5
BR11	7	-41.8	357.0	-53.2	279.6
BR12	7	-11.1	17.5	-26.1	357.1
BR13	7	-29.7	30.9	-54.8	13.4
BR14	11	-73.7	140.1	-85.5	48.9
MW01	6	-46.8	293.4	-85.3	336.3
MW02	6	40.1	114.3	70.7	129.6
MW03	6	35.8	116.2	61.9	147.4
MW04	6	46.5	119.2	72.5	220.4
MW05	7	46.5	107.9	84.1	171.9
MW06	7	26.9	139.4	39.3	189.6
MW07	5	60.5	133.5	45.9	279.9
MW08	4	40.4	145.0	48.6	230.6
MW10	6	52.7	110.9	78.4	269.9
HR01	3	-15.8	263.4	78.6	299.6
HR02	5	-9.6	241.7	87.5	105.8
HR04	4	-11.4	246.6	65.5	7.7
LS02	3	19.8	336.5	-60.8	300.4
LS06	4	31.2	336.8	-61.3	339.1
MP01	7	-16.7	40.1	-52.3	164.0
MP02	8	15.9	347.4	-57.8	57.3
PM01	7	1.1	161.1	36.5	163.2
PM02	6	-15.9	197.9	58.3	217.3
PM03	6	-19.9	224.5	84.9	288.1
PM04	7	-3.9	177.2	69.9	85.2
TP01	16	63.6	340.1	67.3	6.2

N, number of accepted samples. Refer to Figure 2 for sources and locations of sites.

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TABLE 3. Mean Paleomagnetic Poles for the Crescent Formation and Their Implied Rotation and Poleward Displacement

		Uncorrected				Corrected					Poleward
Locality	N	North Latitude	East Longitude	A_{95}	MAD	North Latitude	East Longitude	A_{95}	MAD	Rotation	Displacement
PT/PL	11	59.9	210.9	38.8	42.2	71.6	173.3	22.1	33.3	-17.6 ± 38.8	-2.5 ± 22.3
BR	14	79.6	300.9	19.6	30.5	85.3	218.0	13.0	23.0	7.8 ± 21.9	-2.0 ± 13.3
MW	9	44.8	122.5	8.9	12.9	73.1	209.3	17.1	23.8	-6.3 ± 38.8	-11.6 ± 17.4
This Study	12	25.8	66.8	62.0	38.6	86.4	170.0	16.5	27.3	5.3 ± 26.3	1.1 ± 16.7
All	46	75.5	123.7	18.7	47.5	80.7	192.0	8.0	28.1	0.8 ± 14.4	-3.6 ± 8.5
Expected	----	82.8	170.4	3.0	----	82.8	170.4	3.0	----	----	----

N, number of sites; A_{95}, half-angle of cone of 95% confidence; MAD, maximum angular deviation; PT/PL, Port Townsend/Port Ludlow [*Beck and Engebretson*, 1982; *Warnock*, 1989]; BR, Bremerton [*Beck and Engebretson*, 1982; *Purdy*, 1987]; MW, Mount Walker [*Warnock*, 1989]; Expected, 44-54 Ma cratonal pole of *Diehl et al.*, [1983]. Rotation and poleward displacement calculated from tilt corrected poles. Note that the increase in A_{95} and MAD for the MW locality on correction for tilt is an artifact of the dipole formula; the dispersion of site-mean directions does not change after tilt correction.

Domal uplift probably was caused by the growth of the subduction complex of the Olympic Core terrane. Factors contributing to the uplift may include: isostatic rise of the relatively buoyant sediments of the subduction complex [*Cady*, 1975], upward arching of the down-going Juan de Fuca plate caused by the curved geometry of the trench [*Brandon and Calderwood*, 1990], and/or tectonic thickening during formation of the subduction complex [*Cady*, 1975; *Davis and Hyndman*, 1989]. Tectonic thickening is preferred because it is consistent with models describing the formation of accretionary wedges [*Davis et al.*, 1983]. *Davis and Hyndman* [1989] applied the critically-tapered wedge model [*Davis et al.*, 1983] to the Olympic Mountains. The shallow dip (8° to 10°) of the subducting Juan de Fuca plate [*Crosson and Owens*, 1987] and the large supply of sediment being shed off of the continent due to the rapid rise of the Omineca Crystalline Belt in southeastern British Columbia [*Heller et al.*, 1992], permitted the accretionary prism to rise above sea level [*Davis and Hyndman*, 1989]. Further evidence in support of domal uplift might be the radiating pattern of pencil structures within the eastern Olympic Core [*Tabor and Cady*, 1978b].

Relation to Regional Paleomagnetic Poles

The concordant paleomagnetic pole determined from this study is consistent with the pole calculated from uncorrected directions of the correlative Metchosin Igneous Complex on southern Vancouver Island [*Irving and Massey*, 1990]. One interpretation is that this supports their assumption that the entire exposure of the Metchosin is, on average, horizontal. Consistency also supports the contention [*Clowes et al.*, 1987] that the Metchosin is continuous with the Crescent Formation.

In comparison with other studies of Coast Range basement, the observation of no significant net rotation is anomalous. The

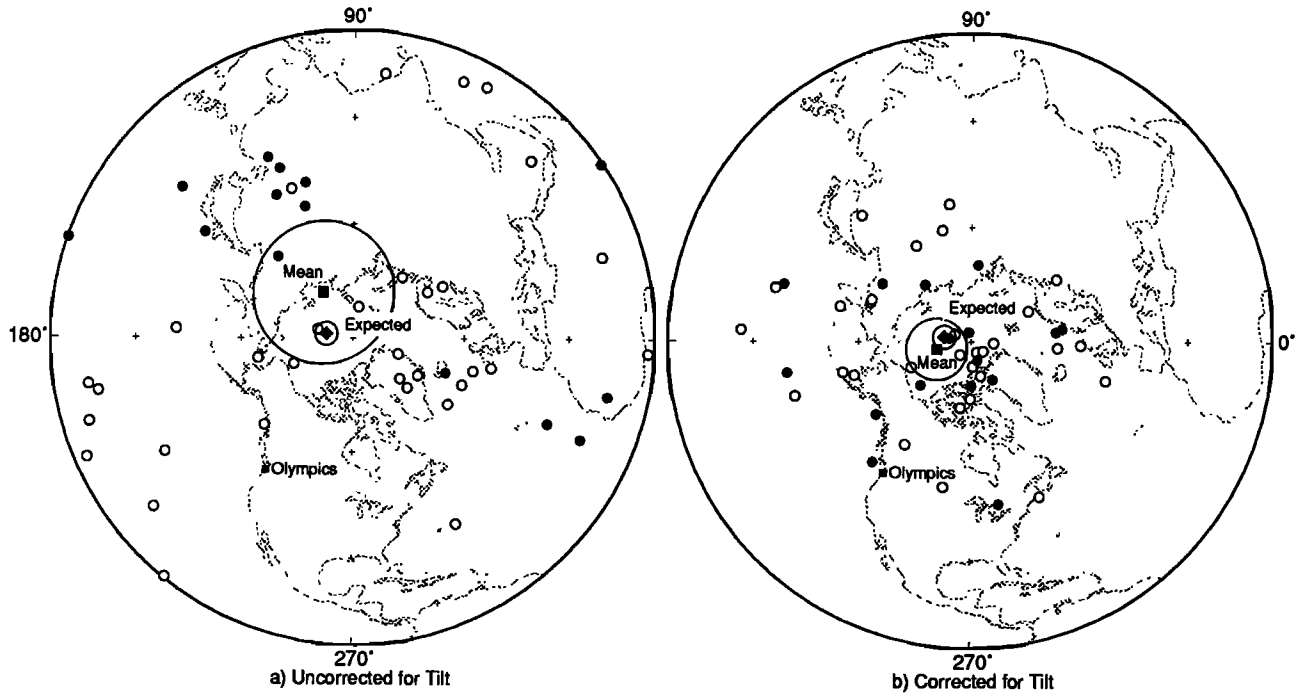


Fig. 7. Equal-area polar projections of site-mean VGPs for the combined data from the Crescent Formation (Table 2) (a) before and (b) after correction for bedding tilt, and their mean (square). The expected 44-54 Ma pole of *Diehl et al.* [1983] (diamond) is also shown. Solid symbols indicate upper hemisphere poles and open symbols represent the antipoles of lower (southern) hemisphere VGPs. Ellipses are A_{95}.

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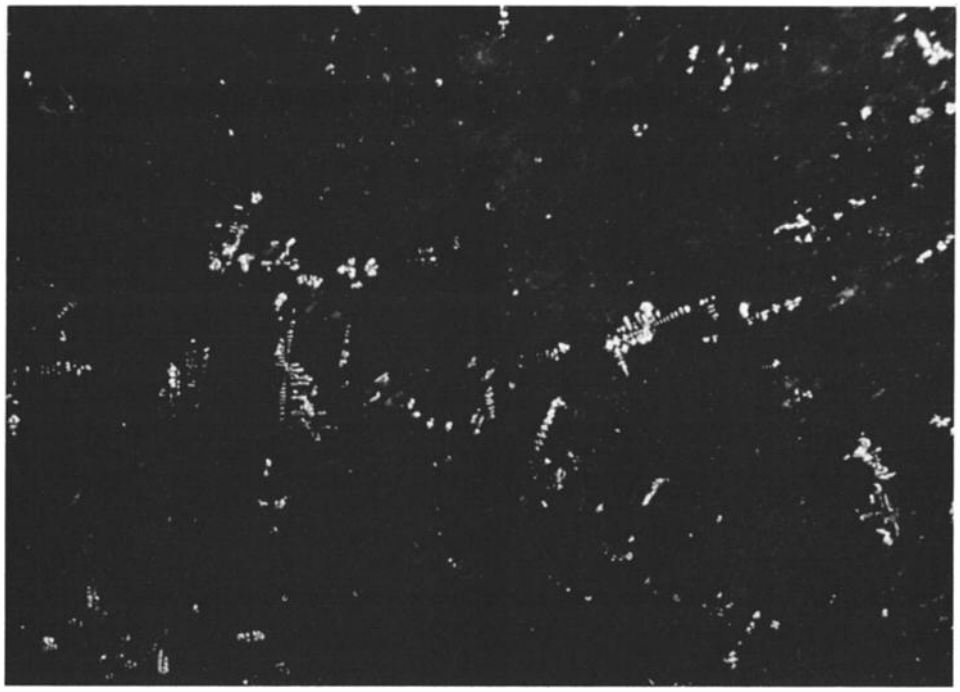
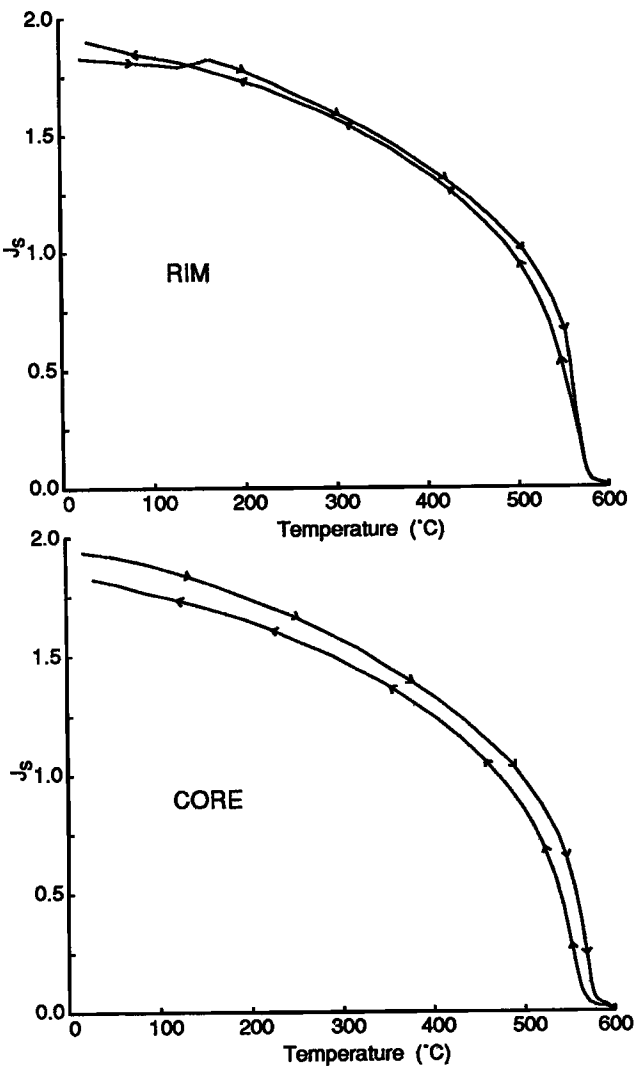


Fig. 8. Photomicrograph of a polished thin section of a pillow core under reflected light. Skeletal growth crystal morphologies (cruciform type) are typical of rapidly quenched titanomagnetite [*Haggerty*, 1991]. The field of view is approximately 0.25 mm wide.



lack of significant rotation could represent a history of approximately equal and opposite rotations [e.g., *Beck and Engebretson*, 1982]. However, such a complex history of rotation would have had to affect, to the same degree, widely spaced localities. Such structural cohesion has not been observed in the Willipa Hills of southwestern Washington where paleomagnetic data and geologic mapping demonstrate that individual structural domains have rotated differentially along closely spaced faults [*Wells and Coe*, 1985] (Figure 1). Nor should such cohesion be expected from our data set given the obvious differential tilting of the fault-bounded localities sampled. Rather, the reason the Crescent Formation shows no rotation may lie in its location against the thick crustal section of Vancouver Island.

How the position of the Crescent Formation explains lack of rotation can be appreciated by reviewing mechanisms that may have driven rotations farther south. *Wells and Heller* [1988] developed a model that describes the rotations of the Coast Ranges in which 60% of the rotation is due to differential extension of the Basin and Range Province inboard of the Coast Ranges [*Simpson and Cox*, 1977; *Magill et al.*, 1981; *Frei et al.*, 1984; *Grommé et al.*, 1986], and 40% is due to nonrigid dextral shear caused by coupling during oblique convergence [*Beck*, 1976, 1980; *Reidel et al.*, 1984; *Sheriff et al.*, 1984]. The former mechanism would not have been effective if the Olympic Mountains were north of the influence of Basin and Range extension [*Wells and Heller*, 1988], but oblique convergence should still have produced some effect. One possible effect is northward translation, suggested by the bias toward anomalously shallow

Fig. 9. Thermomagnetic curves for the rim and core of a single pillow. Specimens were heated in a field of 232.2 mT. The curves are characteristic of essentially pure magnetite with a possible minor amount of pyrrhotite in the rim. Similarity between heating and cooling curves (as shown by arrows) indicates that there were negligible thermally-induced mineralogical changes (courtesy of C.S. Grommé).

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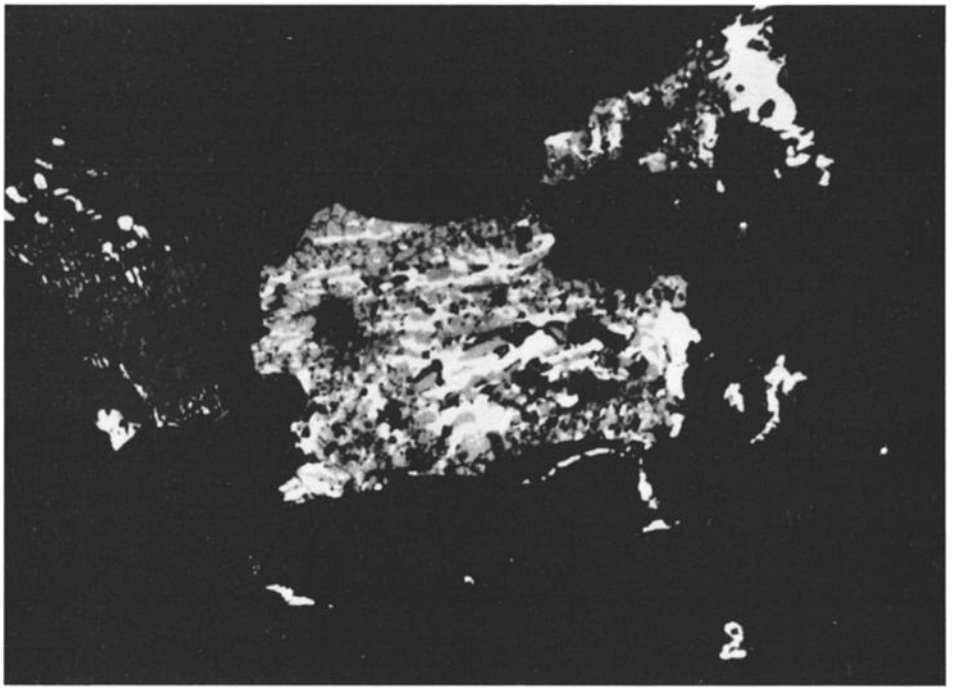


Fig. 10. Photomicrograph of a polished thin section of a subaerial basalt from the upper Crescent Formation showing an oxidation texture of titanohematite [*Haggerty*, 1991] under reflected light. Extreme deuteric oxidation of titanohematite resulted in the exsolution of hematite (white) and pseudobrookite (gray) [*Gromme et al.*, 1969]. The field of view is approximately 0.25 mm wide.

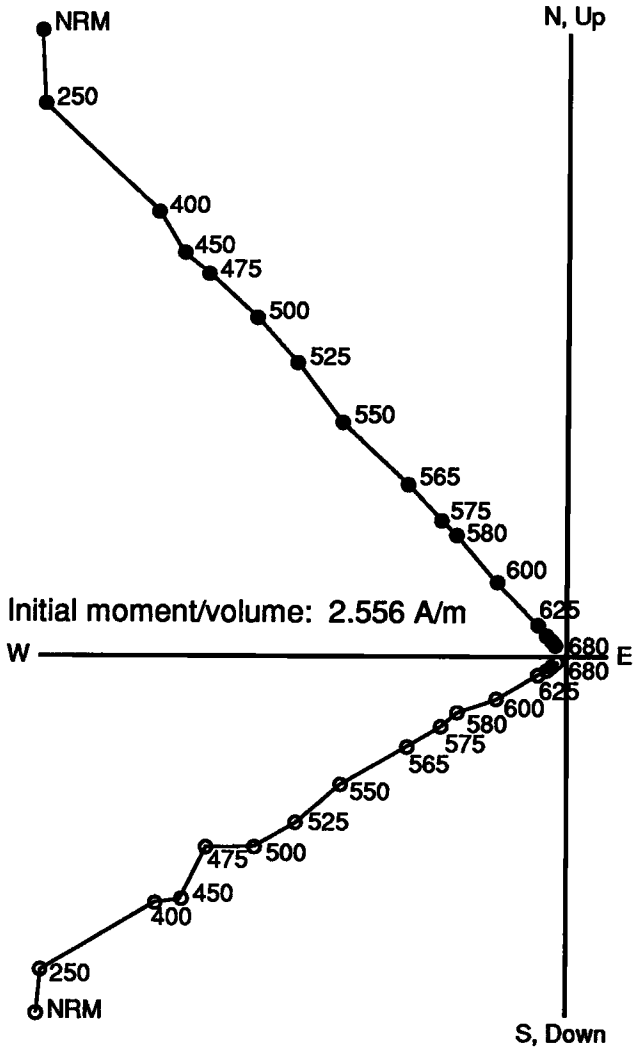


Fig. 11. Orthogonal projection of the thermal demagnetization path for a specimen of subaerial basalt from the Mount Walker locality showing a single component of magnetization that spans the unblocking spectra of both magnetite (below 580°C) and hematite (below 680°C). Solid circles, projection onto horizontal plane; open circles, projection onto vertical west-east plane.

inclinations in other results from the Coast Ranges [*Beck*, 1984]. Although the results from this study show no significant poleward displacement, there may still be geologically important northward (or southward) translation of the northern Crescent Formation within the approximately 800 km confidence limit of the paleomagnetic pole. Another effect would be clockwise rotation if translation decreased continuously eastward in nonrigid crust [*England and Wells*, 1991]. East of the Olympic Mountains, however, all the translation may have been accommodated by strike-slip faults such as the Straight Creek fault [*Vance*, 1992] and one in the Puget Lowlands [*Johnson*, 1984, 1992], both of which were presumably active during or shortly after Crescent volcanism. The reason for this localization of slip may be that thick pre-Tertiary crust of Vancouver Island acted as a rigid block, resisting penetrative deformation. This would reduce the driving mechanism for rotation near Vancouver Island. Evidence that northward translation decreases to the north might be the Clallam syncline (Figure 2). Folds adjacent to the Clallam syncline having roughly east-west trending fold axes [*Snavely and Wells*, 1991] have been shown to affect sediments as young as the Holocene in the Strait of Juan de Fuca (Figure 2) [*Wagner and Tomson*, 1987]. Further evidence might be the northward dipping Leech River thrust fault (Figure 2) on southern Vancouver Island [*Clowes et al.*, 1987].

CONCLUSIONS

Previous paleomagnetic studies of the Eocene submarine pillow basalts of the lower Crescent Formation have failed. Our

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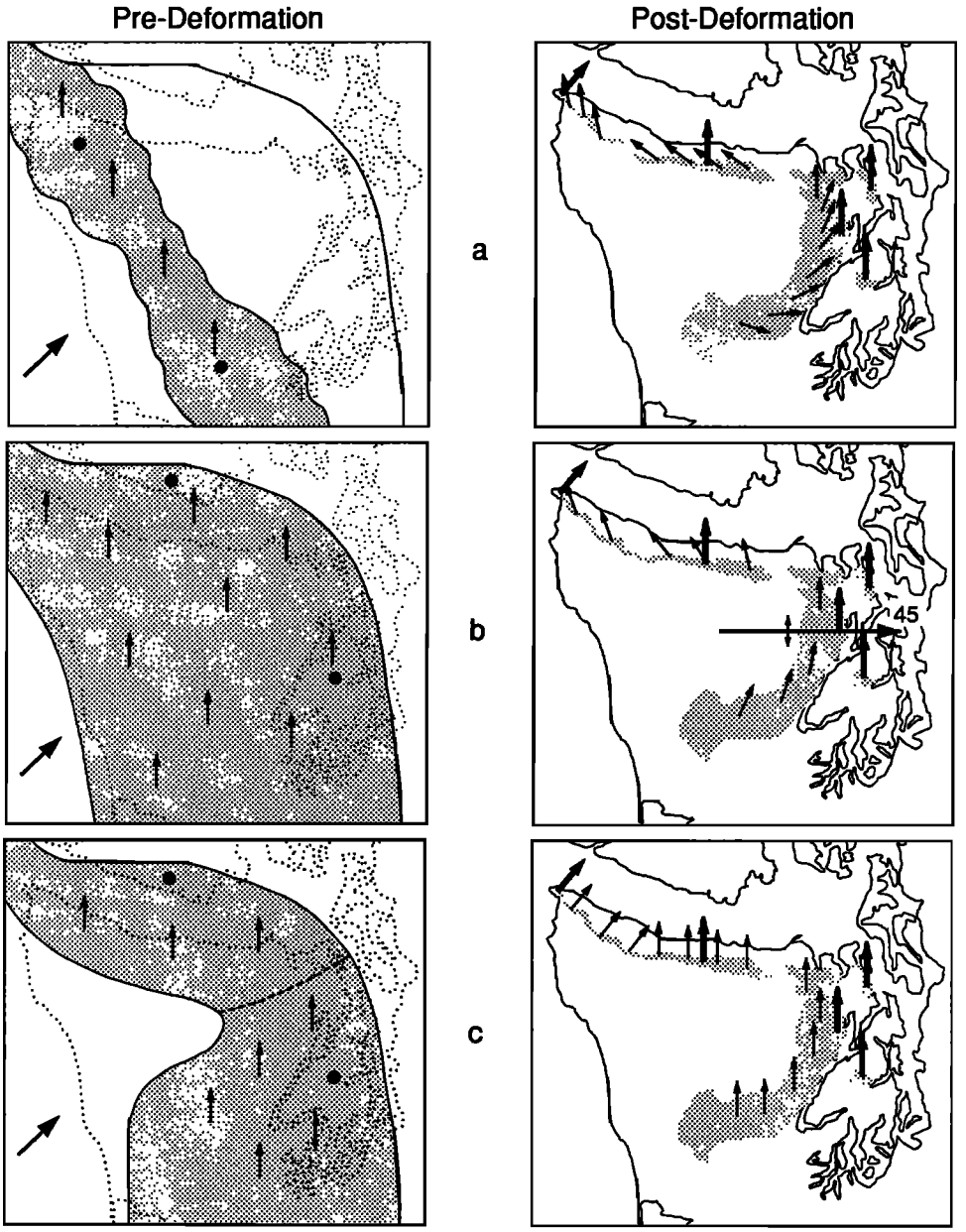


Fig. 12. Observed site-mean directions after correction for local bedding tilt about strike (bold arrows) compared to directions predicted from models proposed for deformation of the Crescent Formation (thin arrows): (a) folding about a vertical axis [*Carey*, 1958]; (b) erosion of an eastward plunging antiform (trend and plunge shown) [*Weaver*, 1937]; (c) erosion of a domelike structure [*Cady*, 1975]. Solid circles in predeformation figures represent possible extrusion centers, and dashed line in Figure 12c represents the possible interfingering of flows from the different vents. Large arrows on the left represent approximate motion of the oceanic plate relative to North America. Note that no paleomagnetic data are available from the southern Olympic Mountains. Secondary components of magnetization from the sedimentary rocks of the northwestern Olympic Peninsula record a later deformational event that produced the approximate 40° clockwise bend of the western tip of the Crescent Formation [*Moyer*, 1985]. This is attributable to oblique convergence of the Farallon and/or Kula plates in conjunction with underthrusting of the subduction complex [*Moyer*, 1985].

early work indicated that the thin, fractured rims of pillows were more likely than their cores to record a stable magnetization. Use of a small-diameter core drill to sample rims selectively contributed greatly to the success of this study.

Three conclusions can be drawn from the paleomagnetic analysis. First, the stable remanent magnetization measured within the Crescent Formation appears to be early, predating significant deformation, and probably is primary. This conclusion is based on rock-magnetic evidence suggesting that the subaerial basalts of the upper Crescent Formation retained their primary remanence. A positive multilimb fold test between exposures of the upper and lower Crescent Formation indicates that the remanence in the lower Crescent was likely acquired early, and while the beds were horizontal.

Second, correction for bedding rotations about strike within four different structural domains produces a circular distribution of VGPs. The simplest model for the deformation that fits the paleomagnetic data is a domelike uplift. The uplift likely was the

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result of tectonic thickening of the subduction complex of the Olympic Core terrane during its formation [*Cady*, 1975; *Davis and Hyndman*, 1989].

Third, the Crescent Formation, where sampled in the north, records no significant net rotation or displacement. This concordance agrees with in situ results from the Metchosin Igneous Complex studied by *Irving and Massey* [1990] and supports their assumption that the entire exposure of the Metchosin is essentially horizontal. Concordance, however, appears anomalous when compared to results from Coast Range exposures to the south. The Olympic Mountains may have been located north of the influence of Basin and Range extension, eliminating one mechanism and more than half of the rotation predicted by the model of *Wells and Heller* [1988]. The other mechanism in this model (*Wells and Heller* [1988]) requires the northward component of plate convergence to drive distributed shear. This mechanism may have been hindered by (1) Vancouver Island acting as a backstop, and (2) residual northward translation, undocumented here, being partitioned into discrete faults east of the Olympic Mountains. Regardless of the mechanisms, the results presented here demonstrate that the northern Olympic Mountains have experienced a deformational history different from correlative Coast Range basement exposures to the south.

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